



IDC216 (Statistics for Scientists) / Monsoon 2025 / Mid Semester Exam 2

Maximum Marks : 20

October 10, 2025

- All questions will be considered in the evaluation.
- Use of calculators is allowed. However, exchange of calculators and use of mobile apps is not permitted.
- Organize your work, in a reasonably neat and coherent way. Work scattered all over the pages without a clear ordering may be missed by the evaluator. Illegible work will not receive credit.
- Mysterious or unsupported answers will not receive credit. A correct answer, unsupported by calculations, explanation, or algebraic work will receive no credit; an incorrect answer supported by substantially correct calculations and explanations *might* still receive partial credit.

1. In a startup incubator, data show that 40% of founders have prior research experience, 35% have prior industry experience, and 20% have both. Determine if the events "Research experience" and "Industry experience" are independent. [2]
2. An experiment evaluated three treatments for Type 2 Diabetes in 699 patients. The three treatments considered were:
 - (met) : treatment with Metformin (which lowers glucose production).
 - (sul) : treatment with Sulfonylureas (which increases insulin production).
 - (lifestyle) : lifestyle intervention.

Each patient had a primary outcome: either *failure* (did not result in blood sugar control) or *success* (resulted in blood sugar control). The results are shown below:

	Failure	Success	Total
lifestyle	109	125	234
met	120	112	232
sulf	90	143	233
Total	319	380	699

- (a) After looking at this data and performing chi-square test, the experimentalist says that she is 95% confident that the choice of treatment affects the outcome. Are you in agreement with this experimentalist? [4]
- (b) A Statistician patient decides that he will start a medicine only when this data provides him 99% confidence that the choice of treatment affects the outcome. After performing chi-square test on this data, did this Statistician start medicine? [2]

3. Six students were ranked based on their performance in three different activities: a Math test, a Hackathon, and a Chess tournament. The ranks are shown in the table below:

Participant	Math Rank	Hackathon Rank	Chess Rank
A	5	3	2
B	2	2	5
C	1	4	6
D	4	1	1
E	3	5	4
F	6	6	3

We denote the Spearman's rank correlation coefficient between Math and Hackathon by $\rho(M, H)$, between Hackathon and Chess by $\rho(H, C)$, and between Math and Chess by $\rho(M, C)$. An AI program suggests that if $\rho(M, H) > 0$ and $\rho(H, C) > 0$ then $\rho(M, C) > 0$. Will you agree with this AI program? [6]

4. At IISER Mohali, the time spent by students on class assignments follows an unknown distribution whose mean is 15 hours and standard deviation is 6 hours. A random sample of 36 students is taken. What is the probability that the sample mean time lies between 14 and 16 hours? [6]

df	$\chi^2_{0.995}$	$\chi^2_{0.990}$	$\chi^2_{0.975}$	$\chi^2_{0.950}$	$\chi^2_{0.900}$	$\chi^2_{0.100}$	$\chi^2_{0.050}$	$\chi^2_{0.025}$	$\chi^2_{0.010}$	$\chi^2_{0.005}$
1	0.000	0.000	0.001	0.004	0.016	2.706	3.841	5.024	6.635	7.879
2	0.010	0.020	0.051	0.103	0.211	4.605	5.991	7.378	9.210	10.597
3	0.072	0.115	0.216	0.352	0.584	6.251	7.815	9.348	11.345	12.838
4	0.207	0.297	0.484	0.711	1.064	7.779	9.488	11.143	13.277	14.860
5	0.412	0.554	0.831	1.145	1.610	9.236	11.070	12.833	15.086	16.750

Table 1: Chi-square distribution table

z	$\Phi(z) = P(Z \leq z)$
-1.0	0.1587
-0.5	0.3085
0.0	0.5000
0.5	0.6915
1.0	0.8413
1.5	0.9332
2.0	0.9772
3.0	0.9987

Table 2: Selected cumulative probabilities for the standard normal distribution